

USING PSO IN CREDIT RATING SYSTEMS

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Quantitative finance has become a challenging field in the past two decades. A significant instance in banking is credit rating (the evaluation of the likelihood of an obligor to default on a loan). This process involves the assessment of credit risk, the assignment of internal risk ratings, and the quantification of default and loss estimates. Within such a system, a key role is played by probability of default rating, in which each obligor is consigned to a certain rating class, or *PD* bucket, based on which a capital charge against credit risk will be computed. This allows to set appropriate conditions on the loan contract based on the client's rating.

Numerous classification techniques have been adopted for credit rating. These techniques include traditional statistical methods (eg. discriminant analysis, linear regression, logit), non-parametric statistical methods (k-nearest neighbor, decision trees), neural networks, etc. The analytical approaches to this problem are based on *k*-means and have some limitations, such as the risk of local optima, the impossibility to add further constraints or to extend the underlying statistical model. Different from traditional search algorithms, evolutionary computation techniques work on a population of potential solutions of the search space. Through cooperation and competition among the potential solutions, these techniques often can find optima more quickly when applied to complex optimization problems. The most commonly used population-based evolutionary computation techniques are motivated from the evolution of nature.

PSO is a population based stochastic optimization technique proposed by Kennedy and Eberhart in 1995 and inspired by the social behavior of bird flocking or fish schooling (Kennedy and Eberhart, 1995). Since then, the PSO has been successfully applied in many research and application areas (including optimization problems) For instance: Neural Network training, Biochemistry, manufacturing, electromagnetism, electrical power, optics and structural optimization.

PSO shares many similarities with evolutionary computation techniques such as GA. The standard PSO algorithm randomly initializes a group of the potential solutions, called particles, and then searches for optima by updating all the particles along a number of iterations. Different from evolutionary computation techniques PSO is motivated from the simulation of social behavior

It is demonstrated that PSO gets better results in a faster, cheaper way compared with other methods. Another reason that PSO is attractive is that there are few parameters to adjust. One version, with slight variations, works well in a wide variety of

applications. Particle swarm optimization has been used for approaches that can be used across a wide range of applications, as well as for specific applications focused on a specific requirement.

Our study introduces a quantitative approach to credit rating based on error minimization of statistical criteria that can be used in the banking industry. In this paper, the performance of PSO (which is a relatively new search heuristic) using credit rating data of an Iranian bank, is demonstrated. Our results show that the choice of the search heuristic to tackle this problem matters and that PSO turns out to be a reasonably apposite approach.

Keywords

Credit rating, PSO, default risk, banking

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